Helwan University

Faculty of Science

El Malahy Smart Park System

Graduation Project Documentation

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Abstract

This project details the design and implementation of the **El Malahy Smart Park System**, an all-in-one digital platform for El Malahy amusement park that makes visiting easier and more fun. The system is developed as a cross-platform solution (Web & Android) that allows visitors to purchase different ticket bundles, then receive QR codes to enter the park. The platform displays all rides and attractions available in the park, along with detailed descriptions and images to help visitors explore and choose what suits them best.

The system learns about each user's age and interests through their account, and an AI chatbot suggests which rides and attractions they might enjoy most. As an interactive and motivational feature, every ride and device in the park has its own unique QR code. When a visitor scans it before riding, they earn loyalty points, allowing the system to accurately track ride popularity and collect instant feedback. These points can be redeemed for rewards, encouraging guests to return.

Staff members have special accounts to scan tickets and see live park stats. After visits, the platform collects feedback through smart surveys to help the park understand what's working well and what needs improvement, making sure every guest's opinion helps shape a better future experience.

Keywords: Amusement Park Management, QR Code System, AI Chatbot, Loyalty Program, Cross-Platform Application.

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1 Introduction

1.1 System Idea (Vision & Scope)

El Malahy Smart Park System is an integrated digital ecosystem designed to transform the traditional amusement park experience into a seamless, tech-driven journey. The system bridges the gap between physical attractions and digital convenience through a cross-platform approach (Web & Android).

1.1.1 Core Value Proposition

- **Convenience:** Digital ticketing and QR-based access eliminate queues and paper tickets.
- **Personalization:** AI-driven recommendations tailored to users' age and interests.
- **Engagement:** A loyalty program that rewards physical interaction with rides.
- **Data-Driven Management:** Real-time analytics and feedback loops for park administrators.

1.2 Problem Statement

Traditional amusement parks often face challenges that diminish the visitor experience and create operational inefficiencies. This project addresses several key problems observed in conventional park environments:

1. **Inefficient Ticketing Processes:** Long queues at ticket booths and reliance on physical paper tickets lead to delays and a frustrating entry experience for visitors.
2. **Lack of Personalization:** Visitors often feel overwhelmed by the number of rides and attractions, with no guidance on what suits their age group, interests, or thrill level. This leads to missed opportunities for enjoyment.
3. **Low Visitor Engagement:** Traditional parks lack interactive features to keep visitors engaged beyond the rides themselves. There is often no incentive to explore all areas of the park.
4. **Limited Data Insights for Management:** Park administrators often lack real-time data on ride popularity, visitor demographics, and crowd flow, making it difficult to optimize operations and allocate resources effectively.
5. **Ineffective Feedback Loops:** Traditional suggestion boxes or manual surveys often result in low response rates and unstructured feedback that is difficult to

analyze for actionable insights.

1.3 Project Aim & Objectives

1.3.1 Overall Aim

To develop an integrated, intelligent digital ecosystem for El Malahy Amusement Park that transforms the visitor journey from ticket purchase to post-visit feedback, enhancing convenience, personalization, and operational efficiency through technology.

1.3.2 Specific Objectives

- **Objective 1:** To digitize and streamline the ticketing process by implementing an online purchase system with QR code generation for secure and rapid park entry.
- **Objective 2:** To enhance visitor engagement and exploration by creating a loyalty program that rewards visitors for interacting with rides and attractions via QR code scanning.
- **Objective 3:** To provide personalized ride recommendations through an AI-driven chatbot that learns from user profiles (age, interests) to suggest suitable attractions.
- **Objective 4:** To empower park staff and management with real-time data, including live park statistics and ride popularity metrics, via dedicated dashboards.
- **Objective 5:** To establish a structured feedback mechanism using smart surveys to collect visitor insights and support data-driven decision-making.

1.4 Project Scope

1.4.1 Included in Scope

- **Cross-Platform Application:** A fully functional website and Android application for visitors to interact with the system.
- **Digital Ticketing Module:** Functionality for users to view, select, and purchase different ticket bundles (e.g., family pass, student pass, VIP) and receive a unique, scannable QR code for entry.
- **AI-Powered Chatbot:** An intelligent chatbot integrated into the app and website capable of recommending rides based on user profile data.
- **Loyalty & Gamification Module:**
  - QR codes on individual rides that visitors can scan.

  - A points system that credits user accounts upon scanning.
  - A rewards store where users can redeem points.
- **Staff Dashboard:** A secure interface for staff to scan visitor QR codes for validation, view live park capacity, and access basic analytics.
- **Admin Dashboard:** A backend interface for park management to monitor ride popularity, view visitor feedback, manage ticket bundles, and generate reports.
- **Smart Surveys:** Automated post-visit surveys that adapt questions based on the visitor's activity (e.g., which rides they scanned).
- **Database Management:** Centralized storage for user data, ticket transactions, scan history, and feedback.

1.4.2 Excluded from Scope (Current Version)

- **Physical Hardware Integration:** The project covers the software logic for QR scanning but does not include integration with physical turnstiles or gates.
- **Advanced Payment Gateways:** A simulated payment gateway is used for the prototype; integration with a live banking API is considered future work.
- **Predictive Analytics:** While the system provides live stats, predictive forecasting of crowd flow based on historical data is excluded.

1.5 Significance of the Project

The El Malahy Smart Park System delivers significant value by addressing the core inefficiencies of traditional amusement parks:

- **Enhanced Visitor Experience:** By eliminating ticket queues, providing personalized ride suggestions, and adding a fun, rewarding layer of gamification, the system turns a simple park visit into a memorable, personalized journey.
- **Increased Operational Efficiency:** Digital ticketing reduces the need for large ticket booth staff. Real-time data allows management to allocate resources where they are needed most, optimizing operational costs.
- **Data-Driven Strategy:** The system empowers park management with actionable insights. Understanding ride popularity, demographics, and visitor feedback allows for targeted marketing and service improvements.
- **Improved Staff Workflow:** The staff dashboard provides a single, mobile-friendly interface for ticket validation and live updates, replacing cumbersome manual checklists.
- **Competitive Advantage:** By embracing AI, cross-platform accessibility, and gamification, El Malahy Park positions itself as a modern, tech-forward destina-

tion, increasing visitor retention through the loyalty program.

2 System Requirements

2.1 Functional Requirements

The system shall perform the following specific functions, categorized by user role:

2.1.1 User & Authentication Management

- The system shall allow users to register, log in, and log out securely.
- The system shall support role-based authentication for Visitor, Staff, and Admin roles.
- The system shall allow users to manage their profile, including age and interest data.

2.1.2 Visitor Specific Functions

- The system shall allow visitors to browse park information and ride descriptions.
- The system shall allow visitors to view and purchase different ticket bundles.
- The system shall generate a unique QR code upon ticket purchase.
- The system shall provide an AI-powered chatbot for personalized ride recommendations.
- The system shall allow visitors to scan QR codes on rides to earn loyalty points.
- The system shall allow visitors to view their points balance and redeem rewards.
- The system shall allow visitors to submit feedback through smart surveys.

2.1.3 Staff Specific Functions

- The system shall allow staff to scan visitor QR codes to validate tickets.
- The system shall provide staff with a dashboard displaying live park statistics.
- The system shall allow staff to view current park capacity.

2.1.4 Admin Specific Functions

- The system shall allow admins to manage ticket bundles (add, edit, delete).
- The system shall allow admins to view detailed analytics on ride popularity.
- The system shall allow admins to view and analyze visitor feedback.
- The system shall allow admins to manage the rewards store.
- The system shall allow admins to generate reports on park performance.

2.2 Non-Functional Requirements

The system shall adhere to the following quality attributes:

2.2.1 Performance

- The system shall provide fast response times for web pages and API calls.
- QR code scanning shall be processed in near real-time with minimal latency.
- Database queries shall be optimized for efficient data retrieval.

2.2.2 Security

- The system shall implement robust authentication and authorization mechanisms.
- All sensitive data transmission shall be secured using HTTPS.
- The system shall prevent unauthorized access based on user roles.
- QR codes shall be encrypted and validated server-side to prevent forgery.

2.2.3 Usability

- The system shall feature an intuitive interface easy to navigate for all age groups.
- The chatbot shall provide clear, concise, and helpful responses.
- The system shall provide clear feedback on the status of operations.

2.2.4 Scalability

- The system architecture shall support high traffic during peak holiday seasons.
- The platform must be accessible 24/7 for ticket bookings.
- The modular design shall allow for future expansion and addition of new features.

2.2.5 Reliability

- The system shall maintain consistent functionality and availability.
- Error handling mechanisms shall be robust to prevent system crashes.
- Data stored in the database shall maintain integrity and consistency.

2.2.6 Maintainability

- The codebase shall be well-structured, modular, and documented.
- Adherence to software design patterns shall promote maintainability.

3 Stakeholders Identification

The El Malahy Smart Park System involves the following key stakeholders:

Stakeholder	**Description & Role**
Visitors (End Users)	Individuals or families who use the web/app to purchase tickets, explore rides, interact with the AI chatbot, and earn loyalty points.
Park Staff	On-ground employees responsible for scanning QR codes, verifying tickets, and managing ride access.
Park Administrators	Decision-makers who monitor live statistics, analyze ride popularity, and review guest feedback to improve operations.
System Developers	The technical team responsible for maintaining the platform, AI integration, and database security.

Table 1: Project Stakeholders

4 Development Methodology: Waterfall Model

For this project, we have adopted the Waterfall Methodology. This linear and sequential approach ensures that each phase of the project is completed and verified before moving on to the next, which is ideal for achieving the structured requirements of a graduation project.

4.1 Why Waterfall?

- **Clear Structure:** Since the project requirements (Ticketing, AI Chatbot, QR Loyalty) are well-defined from the start, Waterfall provides a disciplined path to completion.
- **Documentation-Focused:** Each stage produces specific deliverables, ensuring a comprehensive final report.
- **Milestone Tracking:** It allows the team and supervisors to easily track progress against a fixed timeline.
- **Scope & Time Management:** Given the fixed academic year timeline, Waterfall helps maintain strict control over the project scope, preventing "feature creep" and ensuring all core modules are delivered on schedule.

4.2 Waterfall Phases

Phase	**Activities**
Requirements	Gathering and documenting all functional and non-functional requirements
Design	System architecture, database design, UI/UX mockups
Implementation	Coding the system modules (Web, Android, Backend, AI)
Testing	Unit testing, integration testing, user acceptance testing
Deployment	System deployment and user training
Maintenance	Bug fixes and future enhancements

Table 2: Waterfall Development Phases

5 System Analysis & Design

5.1 System Architecture

The El Malahy Smart Park System is built on a modular, layered architecture to ensure scalability, maintainability, and separation of concerns.

5.1.1 High-Level Architecture Components

Layer	**Components**
Client Layer	Web Browser (React.js), Android App
Application Layer	API Endpoints, Controllers, Services, Repositories, Authentication (JWT)
AI Module	Chatbot Service (Python/Flask), Recommendation Engine
Data Layer	Supabase (PostgreSQL)
External Services	QR Code Library, Simulated Payment Gateway

Table 3: System Architecture Layers

5.1.2 Architectural Layers Description

- **Presentation Layer:** Responsible for the user interface and handling user interactions. Includes web views and mobile interfaces.
- **Business Logic Layer:** Encapsulates core business rules, including point calculation, ticket validation, and recommendation logic.
- **Data Access Layer:** Provides an abstract interface for accessing and managing data stored in the database.
- **Data Layer:** Defines the structure of the application's data and maps to database tables.

5.2 Role-Based Access Control

The system interacts with several distinct types of users, each with specific responsibilities and access privileges:

Role	**Description & Permissions**
Visitor (Unregistered)	Can browse park info and ride descriptions. Cannot purchase tickets or use chatbot.
Registered User	Can manage profile, purchase tickets, use chatbot, scan QR codes, earn points, redeem rewards. Access limited to own data.
Park Staff	Can scan visitor tickets, view live park stats (attendance, ride activity).
Park Admin	Full system access. Manage ticket bundles, view analytics, manage user accounts, generate reports.

Table 4: Role-Based Access Control

5.3 Functional Design

5.3.1 Key Use Cases

Use Case	**Description**
Purchase Ticket Bundle	User selects a bundle, pays, and receives a unique QR code.
Get AI Recommendation	User asks chatbot for suggestions; system returns ride options.
Scan Ride QR Code	User scans code at a ride; system credits points and logs scan.
Validate Entry Ticket (Staff)	Staff scans visitor QR code; system verifies validity.
View Live Park Stats	Staff/Admin accesses dashboard to see real-time data.
Manage Rewards (Admin)	Admin adds or removes rewards in the points store.

Table 5: Key Use Cases

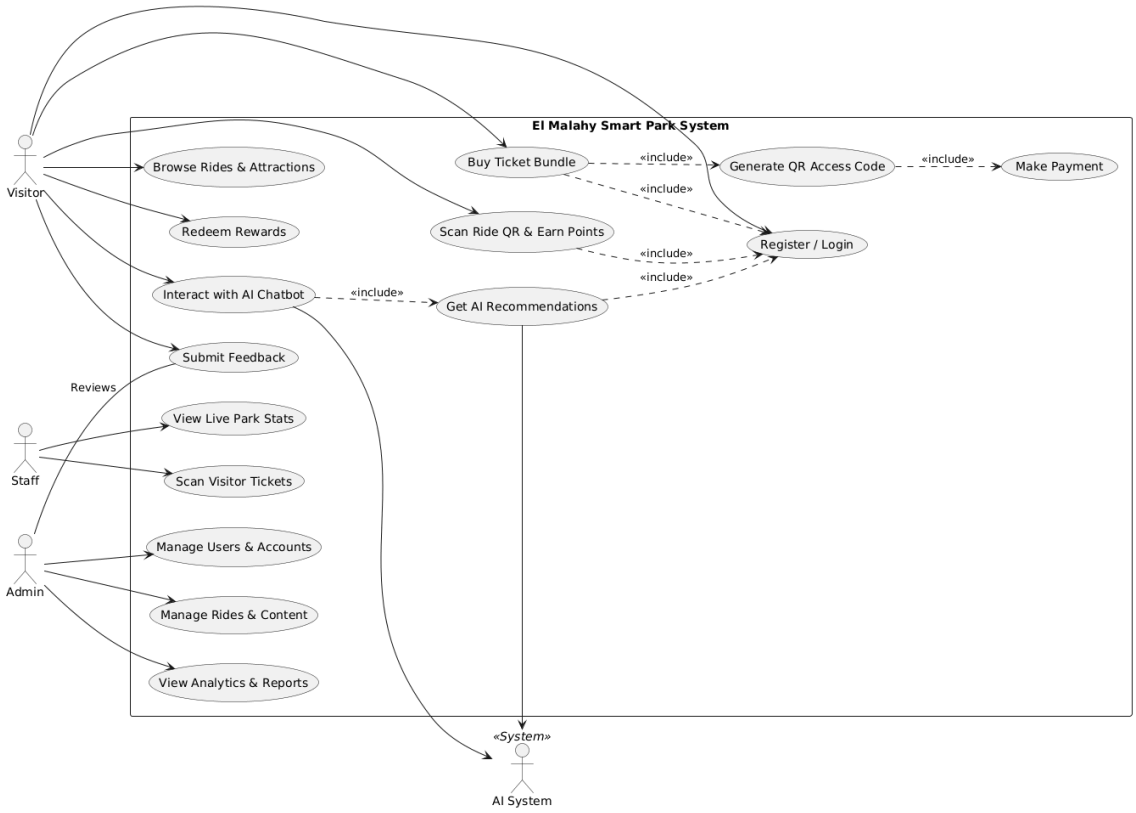

Figure 1: Use Case Diagram for El Malahy Smart Park System

5.4 Database Design

5.4.1 Database Entities

The following entities form the core of the database schema:

Entity	**Key Attributes**
Users	UserID (PK), Name, Email, PasswordHash, Age, Interests, Role
Tickets	TicketID (PK), UserID (FK), BundleType, PurchaseDate, QRCodeData, IsUsed
Rides	RideID (PK), Name, Description, MinAge, MaxAge, ThrillLevel, ImageURL, QRCodeValue
Scans	ScanID (PK), UserID (FK), RideID (FK), ScanTimestamp, PointsEarned
Rewards	RewardID (PK), Name, Description, PointsRequired, Stock
Redemptions	RedemptionID (PK), UserID (FK), RewardID (FK), RedemptionDate
Feedback	FeedbackID (PK), UserID (FK), VisitDate, Rating, Comments

Table 6: Database Entities

5.5 UI/UX Design Principles

The User Interface of the El Malahy Smart Park System is designed according to the following principles:

- **User-Centric Design:** Interface designed to be intuitive for all age groups, with large buttons and clear instructions for QR scanning.
- **Consistency:** A unified design language (colors, fonts, icons) is applied across the website and Android app.
- **Gamification Elements:** Progress bars for points, badges for achievements, and a visually appealing rewards store to motivate users.
- **Responsiveness:** The web interface is fully responsive, adapting seamlessly from desktop monitors to mobile screens.
- **Feedback and Error Handling:** The system provides clear and immediate visual feedback regarding user actions.

6 Implementation

6.1 Technologies & Tools Used

Component	**Technologies**
Backend API	Node.js
Web Frontend	React.js, HTML5, CSS3, JavaScript
Mobile App	Android (Kotlin/Java)
AI Chatbot	Python, Flask, NLTK
Database	Supabase (PostgreSQL)
QR Code	QRCoder Library
Authentication	JWT (JSON Web Tokens)
Development Tools	Visual Studio, VS Code, Android Studio

Table 7: Technologies and Tools Used

6.2 Module Implementation Overview

6.2.1 QR Code Generation & Validation

Upon ticket purchase, the system generates a unique, encrypted token and encodes it into a QR code. The token is stored in the database. When a staff member scans the code, the system validates the token against the database, ensures it hasn't been used, and updates its status. This ensures secure and fraud-resistant entry.

6.2.2 Loyalty Points Logic

When a user scans a ride's QR code, a transaction is recorded. The system checks for duplicates (e.g., preventing multiple scans of the same ride in a short period) and then credits a predefined number of points to the user's account. The points balance is updated in real-time.

6.2.3 AI Recommendation Algorithm

The chatbot uses a hybrid approach for recommendations:

1. Extracts user attributes (age, interests) from the user profile.
2. Matches attributes against a database of rides:
   - If age < 12, suggest rides from a "Kids" category.
   - If interests contain "High Speed," suggest roller coasters.

3. If no clear match, recommend the top 5 most popular rides from the last week.

6.3 Challenges Encountered & Solutions

6.3.1 Challenge 1: Ensuring QR Code Uniqueness and Security

- **Problem:** Preventing two tickets from having the same QR code or preventing code forgery.
- **Solution:** A combination of UUIDs (Universally Unique Identifiers) and server-side secret was used to generate tokens. The QR code contains only a pointer to the token, while validation is strictly server-side, preventing client-side tampering.

6.3.2 Challenge 2: Managing High Concurrency for Scans

- **Problem:** During peak hours, many users might scan ride QR codes simultaneously, leading to potential database overload.
- **Solution:** Implemented asynchronous logging for scans. Scans were written to a message queue and processed in batches, ensuring the app remained responsive and preventing database deadlocks.

6.3.3 Challenge 3: Building AI Recommendation Engine Without Historical Data

- **Problem:** In the initial system launch, there was no historical data on user preferences to train a collaborative filtering model.
- **Solution:** The initial recommendation engine was built as a hybrid system using content-based filtering (matching user's stated age and interests to ride attributes) and a popularity-based fallback (showing the most scanned rides of the week).

6.3.4 Challenge 4: Cross-Platform Consistency

- **Problem:** Maintaining consistent behavior and appearance across the web application and Android app.
- **Solution:** A centralized RESTful API was used to serve data to both platforms, ensuring business logic consistency. A shared design system was documented and followed by both development teams.

7 Testing & Evaluation

7.1 Testing Methodology

A multi-faceted testing approach was adopted to ensure system quality:

- **Unit Testing:** Individual functions (point calculation, QR generation, authentication) were tested in isolation.
- **Integration Testing:** The interaction between the backend API and database was tested. Endpoints for ticket purchases and QR code validation were rigorously tested.
- **System Testing:** End-to-end scenarios were simulated, covering the full user journey from registration to reward redemption.
- **User Acceptance Testing (UAT):** A small group of users (visitors) and staff tested the system in a simulated environment. Feedback was collected and used for refinement.

7.2 Test Cases & Results

Test ID	Scenario Description	Expected Outcome	Status
TC-01	User purchases a ticket bundle	System generates unique QR code; user account updated	**Pass**
TC-02	Staff scans valid, unused QR code	System marks ticket as used; success message displayed	**Pass**
TC-03	Staff scans already used QR code	System displays "Ticket already used" error	**Pass**
TC-04	User scans QR code on a ride	System credits loyalty points; logs the scan	**Pass**
TC-05	User asks "What rides can my 10-year-old go on?"	Chatbot returns rides with appropriate age limits	**Pass**
TC-06	Staff logs into dashboard for live stats	Dashboard displays current attendance and top rides	**Pass**
TC-07	Admin adds a new reward to the store	New reward appears in user rewards store	**Pass**
TC-08	User redeems points for a reward	Points deducted; redemption recorded in database	**Pass**

Table 8: Selected Test Cases and Results

7.3 Performance Evaluation

- **Web Application Responsiveness:** Page load times for standard operations were generally quick in development environment.
- **QR Scanning Latency:** QR code validation processed in under 2 seconds under normal conditions.
- **Chatbot Response Time:** Responses for rule-based queries were near-instant; more complex queries showed slightly higher latency based on external API response times.
- **Database Performance:** Optimized queries ensured efficient data retrieval even with multiple concurrent users.

8 Results & Discussion

8.1 Achieved Functionalities

The completed system successfully delivers all core modules:

- **Cross-Platform Application:** Fully functional website and Android app for ticket purchase and exploration.
- **Digital Ticketing:** Functional QR code-based entry system with secure validation.
- **AI-Powered Chatbot:** Interactive chatbot providing personalized ride suggestions based on user demographics.
- **Loyalty Program:** Points accumulation via ride scanning with redeemable rewards.
- **Staff Dashboard:** Interface for ticket validation and live park statistics.
- **Admin Dashboard:** Backend for management of tickets, rewards, and analytics.
- **Smart Surveys:** Automated post-visit feedback collection.

8.2 Project Objectives Achievement

Objective	**Achievement Status**
Digital Ticketing	Successfully achieved; eliminates physical ticket booths
Visitor Engagement	Gamification effective; users actively sought rides to scan
Personalized Recommendations	Hybrid recommendation engine guided users to suitable rides
Real-time Data	Staff dashboard provided valuable live insights for management
Structured Feedback	Smart surveys provided clear, actionable data

Table 9: Objectives Achievement Summary

8.3 Project Limitations

Despite successful implementation, the current version has certain limitations:

- **Scalability Testing:** The system was tested with simulated load, but real-world peak-hour performance with hundreds of concurrent users is not fully validated.
- **AI Model Complexity:** The current recommendation engine is rule-based and does not yet leverage deep learning for advanced personalization.

- **Hardware Integration:** The project did not integrate with physical turnstiles or payment terminals, remaining a software prototype.
- **Payment Gateway:** A simulated payment system was used; live payment integration requires additional security and compliance measures.

9 Conclusion & Future Work

9.1 Conclusion

The El Malahy Smart Park System successfully modernizes the traditional amusement park model by integrating digital convenience, AI personalization, and gamification. By addressing the problems of inefficient ticketing, lack of engagement, and limited data insights, the project provides a comprehensive solution that enhances the visitor journey and empowers park management.

9.2 Key Contributions

- Development of a unified cross-platform system for park management.
- Integration of AI-driven personalized recommendations.
- Implementation of a gamified loyalty program encouraging visitor engagement.
- Provision of real-time analytics for staff and administrators.
- Establishment of structured feedback mechanisms for continuous improvement.

9.3 Future Work

The following enhancements are identified for future versions:

- **Integration with Physical Hardware:** Connect software with electronic turnstiles for fully automated entry and exit.
- **Live Payment Gateway Integration:** Replace simulated payment with live integration (Stripe, PayPal) for real financial transactions.
- **Advanced Predictive Analytics:** Use collected scan and sales data to build machine learning models for crowd prediction and wait time estimation.
- **Augmented Reality (AR) Features:** Integrate AR to show ride information or virtual characters by scanning park landmarks.
- **Push Notifications:** Send personalized offers, wait time updates, and reward alerts to users' mobile devices.
- **Voice Integration for Chatbot:** Implement voice input/output capabilities for enhanced accessibility.
- **Multi-language Support:** Add support for multiple languages to accommodate diverse visitors.

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